MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING, 8 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 79.1, 79.2, 80.1 AND 80.2

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SHIPS).

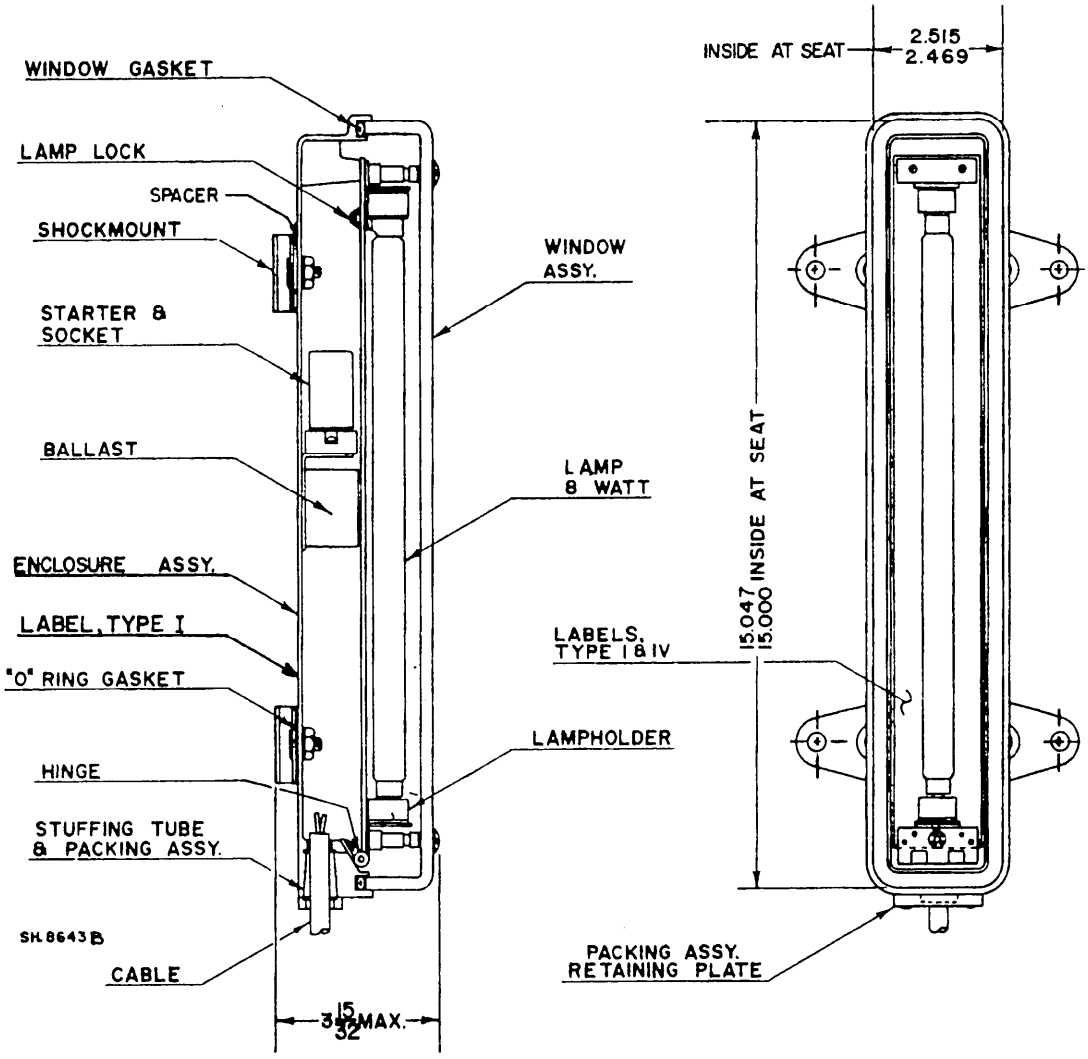


FIGURE 1. Typical view of Symbols 79.1 and 80.1.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

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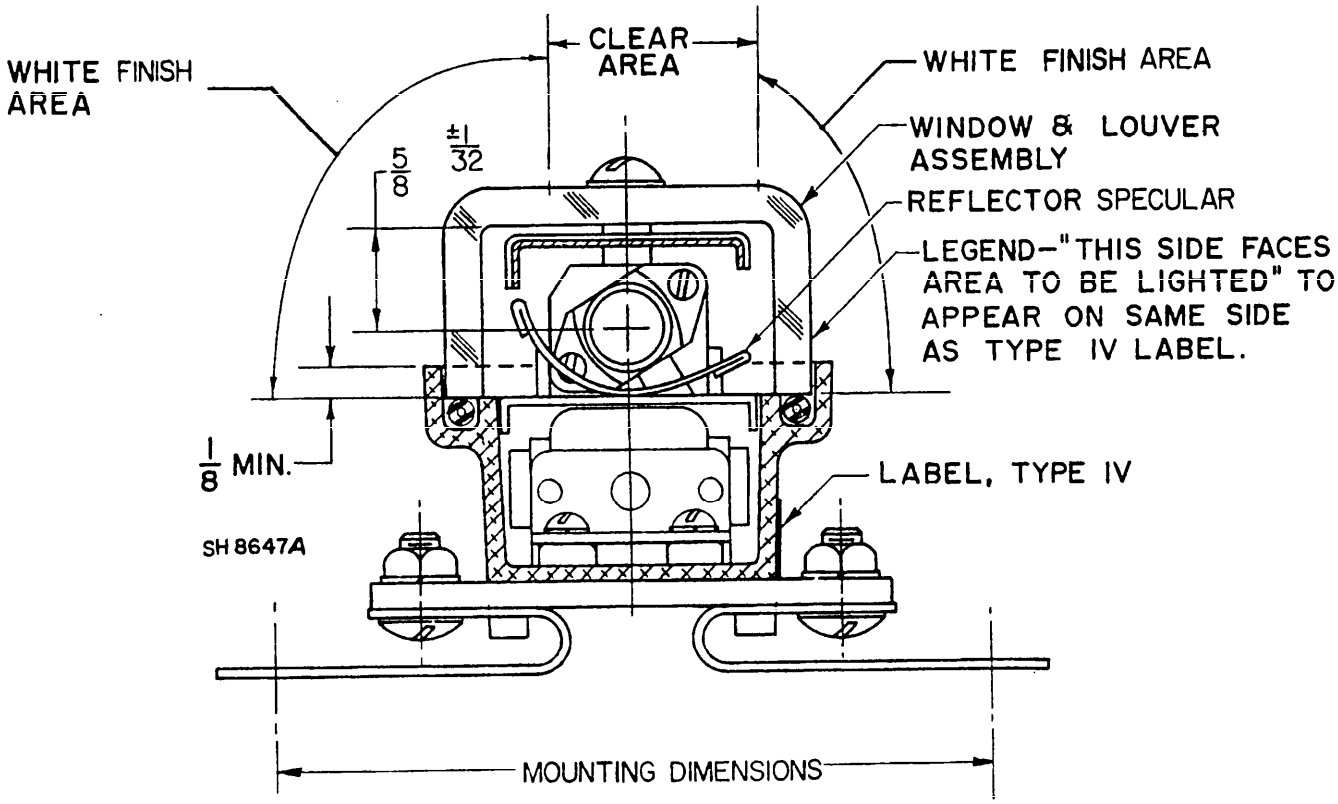


FIGURE 2. Typical section for Symbol 79.1.

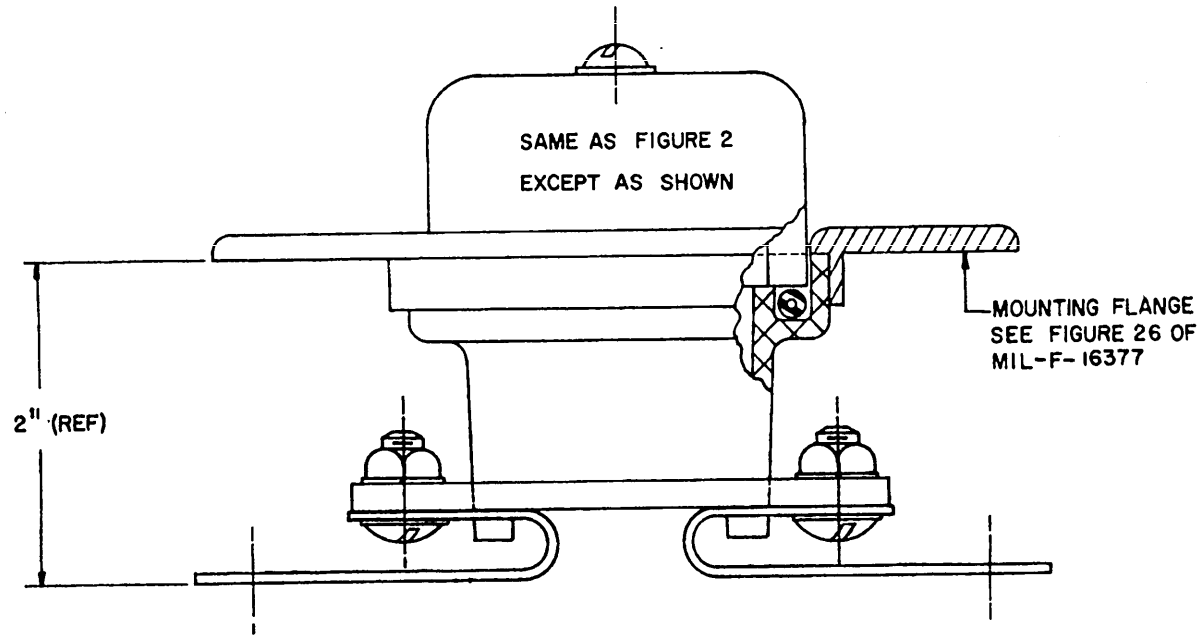


FIGURE 3. Typical section for Symbol 79.2.

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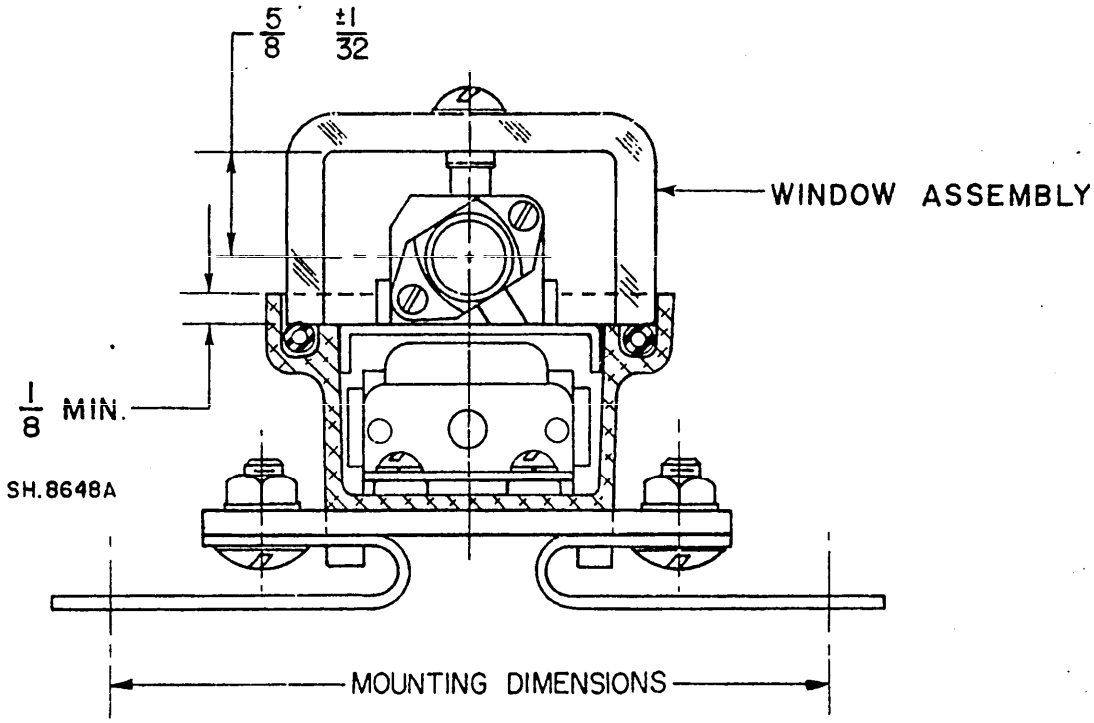
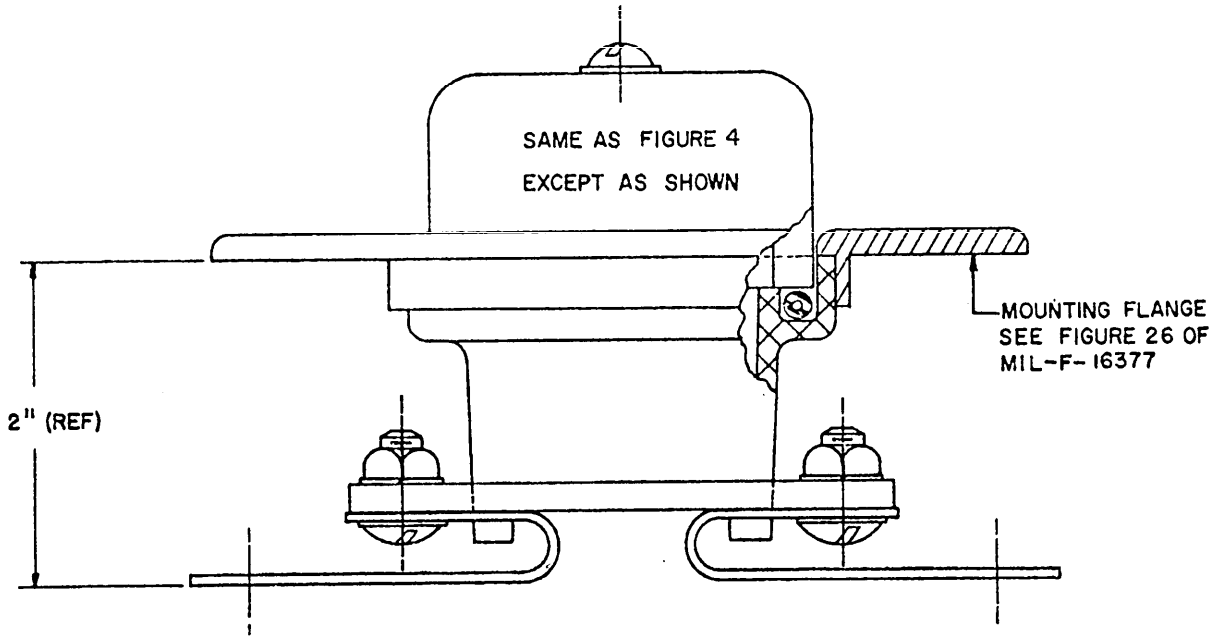


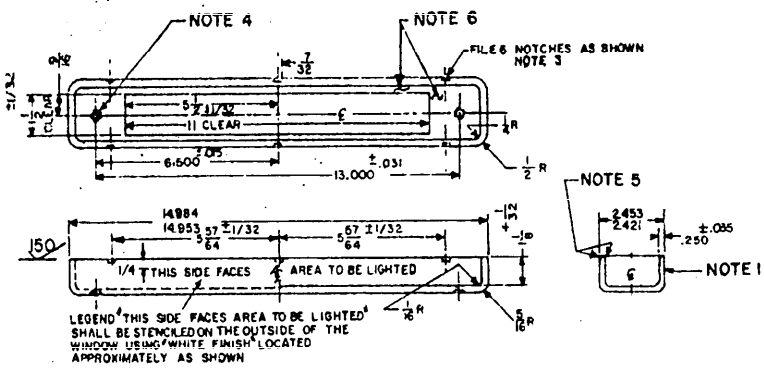
FIGURE 4. Typical section for Symbol 80.1.



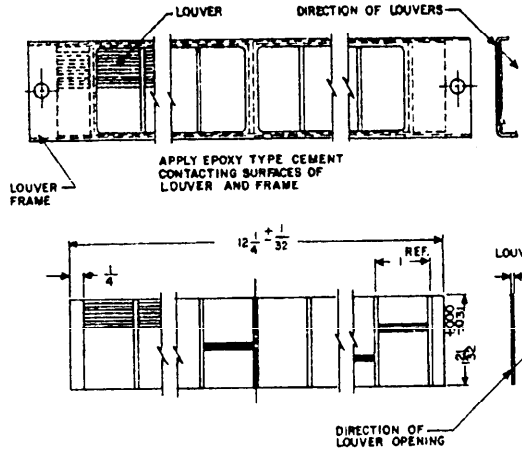
SH 8932 A

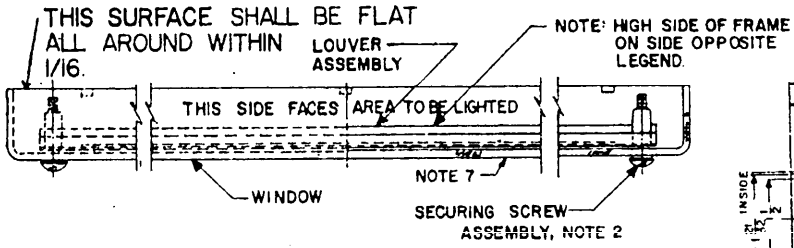
FIGURE 5. Typical section for Symbol 80.2.

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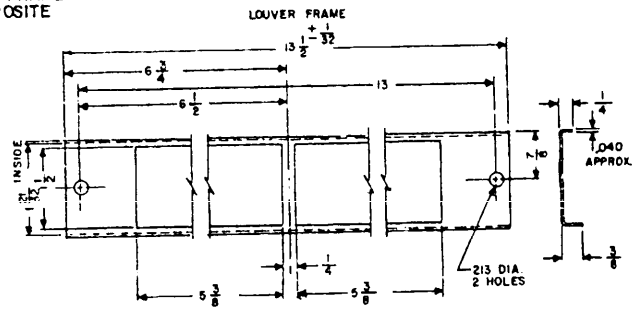


WINDOW DETAILS





WINDOW ASSEMBLY



LOUVER ASSEMBLY AND DETAILS

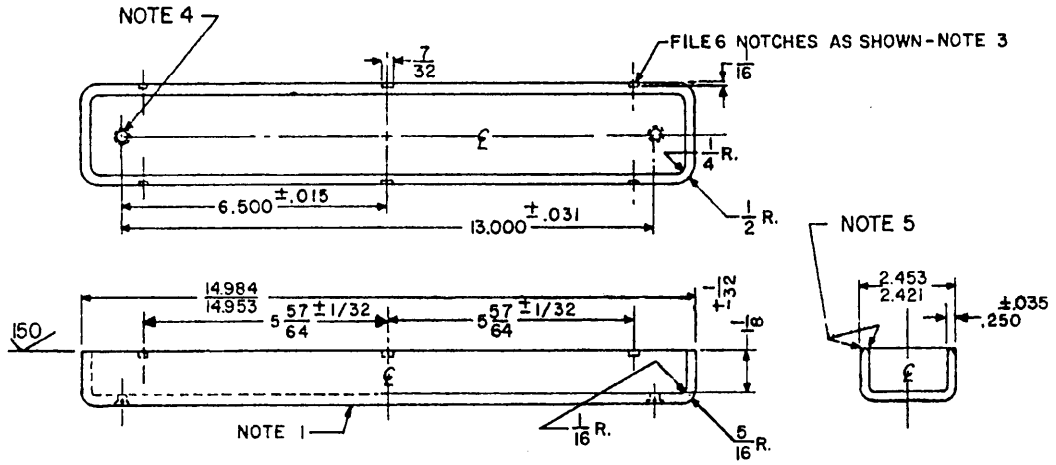
SH 8935B

NOTES:

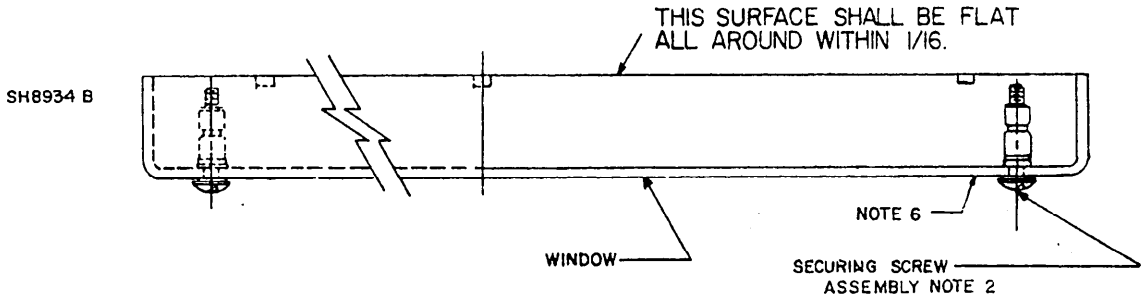
1. Material shall be 1/4 inch thick, clear acrylic plastic, type CLEAR-1 of MIL-P-24191.
2. Two long window securing screw assemblies shall be installed in accordance with 3.7.4.3 of MIL-F-16377 except that the louver frame shall be inserted between leather washer and captive bushing.
3. Notches shall be filed as shown in accordance with 3.7.4.4 of MIL-F-16377.
4. Drill 0.191 diameter hole and counterbore opposite side 0.323 diameter, 0.050 deep. Break sharp edges on the inside surface on all holes.
5. Break sharp edges 1/64 of an inch around entire window.
6. The inside surface shall be finished white as shown in accordance with 3.8.4.1 of MIL-F-16377.
7. When required, onet ype VI label shall be applied on the outside surface in accordance with 3.7.4.3.
8. The louver assembly shall be finished white in accordance with 3.8.4.2 of MIL-F-16377.

FIGURE 6. Window assembly, clear and white, directional, for Symbols 79.1 and 79.2.

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WINDOW DETAILS

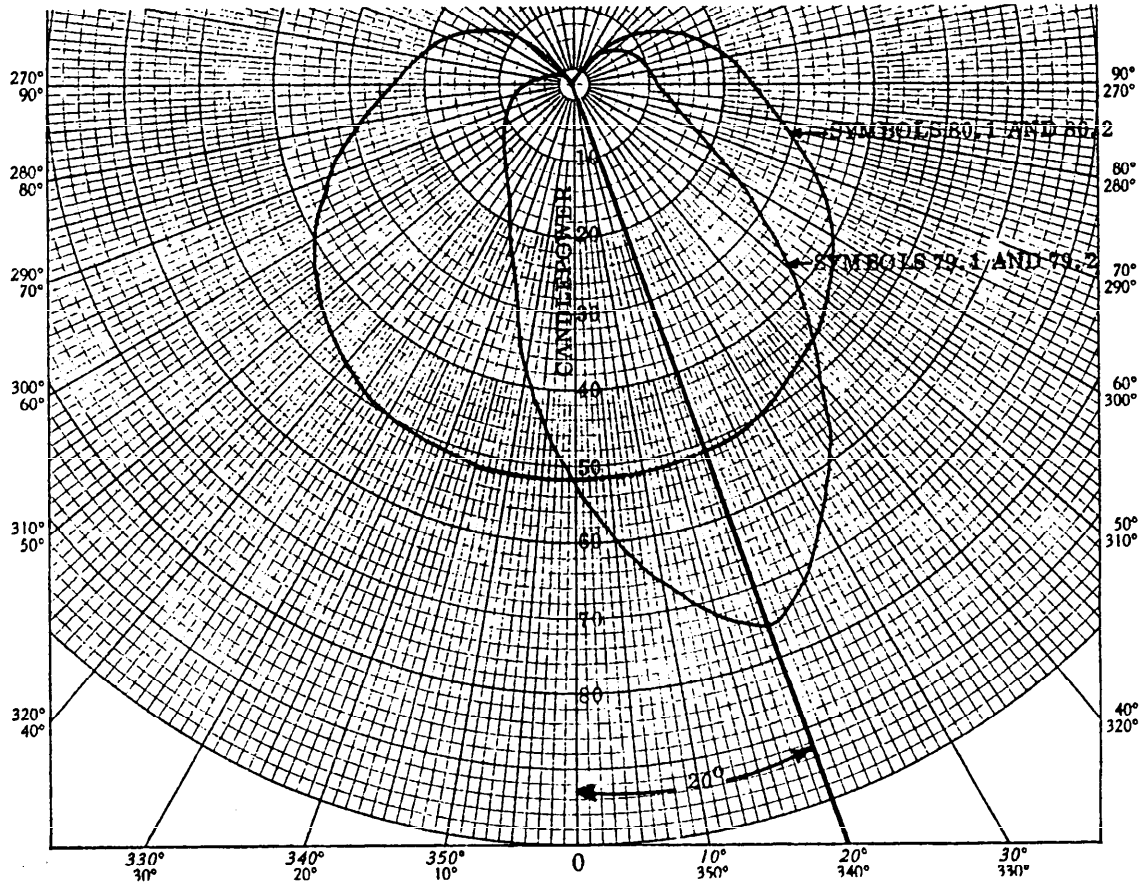


NOTES:

1. Material shall be 1/4 inch thick, white translucent acrylic plastic, Type WTLS-4, Grade G of MIL-P-24191.
2. Two long window securing screw assemblies shall be installed in accordance with 3.7.4.3 of MIL-F-16377 except that two leather washers shall be installed.
3. Notches shall be filed as shown in accordance with 3.7.4.4 of MIL-F-16377.
4. Drill 0.191 diameter hole and counterbore opposite side 0.323 diameter, 0.050 deep. Break sharp edges on the inside surface of all holes.
5. Break sharp edges 1/64 of an inch around entire window.
6. When required, one type VI label shall be applied on the outside surface in accordance with 3.7.4.3 of MIL-F-16377.

FIGURE 7. Window assembly, white, diffusing, for Symbols 80.1 and 80.2.

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SH 8646

*ANGLE	SYMBOL NUMBER	
	79.1 and 79.2	80.1 and 80.2
0	76	53
5	72	52
10	66	52
15	60	52
20	52	51
25	45	51
30	38	50
35	29	49
40	26	47
45	20	46
50	18	44
55	15	42
60	13	40

*ANGLE	SYMBOL NUMBER	
	79.1 and 79.2	80.1 and 80.2
65	12	37
70	11	35
75	11	31
80	11	29
85	10	25
90	10	24
95	9	22
100	9	21
105	8	20
110	6	17
115	6	16
120	5	14
125	4	11

* Measured at an angle of 20° from vertical for Symbols 79.1 and 79.2 only.

Photometric test data:

Distance: 10 feet.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp correction: Distribution curves based on 380 lumens per lamp.

FIGURE 8. Candlepower distribution curves.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 2

SYMBOL NUMBERS:
79.1: Complete lighting fixture assembly, standard mounting with one 8-watt cool white fluorescent lamps and clear and white window assembly, directional.
79.2: Complete lighting fixture assembly, flush mounting, with one 8-watt cool white fluorescent lamps and clear and white window assembly, directional.
80.1: Complete lighting fixture assembly, standard mounting, with one 8-watt cool white fluorescent lamps and white diffusing window assembly.
80.2: Complete lighting fixture assembly, flush mounting with one 8-watt cool white fluorescent lamps and white diffusing window assembly.

DESIGN:
Electrical rating: 8 watts, 120 volts, 60 Hertz.
Dimensions and configurations: See figures 1 through 5 and figure 10 of MIL-F-16377.
Photometric data: See figure 8.
Lampholder spacing: See figure 21 of MIL-F-16377.
Wiring diagram: See figure 22 of MIL-F-16377.
Mounting dimensions:
Symbols 79.1 and 80.1: See figure 25 of MIL-F-16377.
Symbols 79.2 and 80.2: See figure 26 of MIL-F-16377.
Window assembly:
Symbols 79.1 and 79.2: See figure 6.
Symbols 80.1 and 80.2: See figure 7.
Enclosure: Watertight. See 3.5.1] of MIL-F-16377. The enclosure shall be made of 3/32 inch minimum thickness cast aluminum.
Cable: 10 feet of type SJO or SJT, AWG size 16, two conductor (white and black) in accordance with J-C-580.
Weight:
Symbols 79.1 and 80.1: 4 lbs 3 oz. maximum.
Symbols 79.2 and 80.2: 5 lbs 2 ox. maximum.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figures 1 and 2:

(a) Two type I.
(b) One type IV for Symbols 79.1 and 79.2 only. Apply one type IV label as shown on figure 2 bearing legend "THIS SIDE FACES AREA TO BE LIGHTED."
(c) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table 1 and window assemblies of table IV by name and stock number only.

Standard Parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE:

First article inspection:
Complete fixtures: See table II.
Window assemblies: See table III.
Quality conformance: Same as the first article inspection.

Inspection of product for delivery:
Examination: See MIL-F-16377.
Examination for preparation of delivery: MIL-F-16377.
Quality conformance tests: See MIL-F-16377.

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.

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TABLE I. Standard parts.

Name	Specification or Military part number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/3	6240-00-299-5546	Industry number F8T5/CW
Lamplock	M16377/40-001	6250-00-399-7039	
Lampholder	MIL-L-970/10	6250-00-377-8590	
Startersocket	MIL-L-970/13	6250-00-295-2738	
Starter	8W, type III of W-S-755	6250-00-299-5962	Commercial Designation:FS5
Ballast	M16377/45-001	6250-00-295-2735	
Shockmount	M16377/41-010	6210-00-973-1748	See 3.5.16 of MIL-F-16377
Red filter tube	M16377/42-001	6210-00-399-6297	Not furnished see 3.7.7 of MIL-F-16377.

TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

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TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

TABLE IV. Part number and general information.

Military part number M16377/14	Item name	National Stock Number (NSN)	Formerly covered by
79.1	Complete fixture, Symbol 79.1	6210-00-578-7703	MS16673-79
79.2	Complete fixture, Symbol 79.2		
80.1	Complete fixture, Symbol 80.1	6210-00-892-1715	MS16673-80
80.2	Complete fixture, Symbol 80.2		
001	Window assembly, directional	6210-00-553-6793	MS16891-61
002	Window assembly, white diffusing	6210-00-647-2276	MS16891-66

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6210-N560)

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